

Dinor Nallbani

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EDUCATION

University of Massachusetts Amherst

Expected May 2026

B.S. in Mechanical Engineering, Industrial Engineering

GPA: 3.9

- Commonwealth Honors College, Minor in Engineering Management, Dean's List (All Semesters)
- **Coursework:** Design of Mechanical Systems, Dynamics, Fluid Mechanics, Heat Transfer, Strength of Materials

EXPERIENCE

Undergraduate Research Assistant

Sept. 2025 – Present

University of Massachusetts Amherst — Honors Thesis

Amherst, MA

- Researching wind turbine aeroelasticity, focusing on tower 2nd mode frequency and damping response
- Applying the Lagrange method to derive the equations of motion using a symbolic computation framework
- Analyzing the systems with eigenvalue analysis to characterize turbine stability

Smart Building Specialist Intern

May 2025 – Aug. 2025

Siemens

Canton, MA

- Integrated and managed HVAC systems in 5 locations using Siemens' Desigo CC building management system
- Boosted energy efficiency 30% by programming and optimizing logic for over 45 building automation controllers.
- Improving system reliability by diagnosing and resolving issues related to field devices, sensors, and actuators

Mechanical Engineering Student Researcher

Sept. 2024 – May 2025

UMass Amherst, National Renewable Energy Lab.

Amherst, MA

- Collaborated with NREL to evaluate added-mass loads on floating marine hydrokinetic turbines
- Developed a NACA-profile hydrofoil model and mounting mechanism in Onshape for wave-current tank testing
- Sourced required parts and manufactured the foil + mounting mechanism, and a flow straightener in machine shop

UMass Robotics Team

Sept. 2024 – Present

Mechanical Subteam Lead (Humanoid & Research Teams)

Amherst, MA

- Leading 15-member team developing 12-DOF quadruped; managed system integration and technical milestones
- Designing quadruped structure in Onshape, optimizing mass via FEA to meet 25 lb weight constraint
- Engineered 6-DOF humanoid arm cable-driven system, synthesizing dynamics research to optimize design
- Designed humanoid upper arm in Onshape, optimizing for subsystem integration and manufacturability

UMass ASME Mechatronics

Sept. 2024 – Present

Project Engineer (Lunabotics & IAM3D)

- Engineered 6063-T6 Aluminum chassis in SolidWorks, meeting 80kg mass and 1.5x0.75m volume constraints
- Designed 25cm PA6-CF 3D-printed wheels and integrated drivetrain to optimize center of mass
- Led mechanical spatial layout for regolith mining, coordinating subsystem integration with excavation team
- Developed FPV drone using FEA mass optimization and engineered a compliant grabber with 400-cycle fatigue life

R4 Solutions

Summer 2024

Mechanical Engineering Intern

- Engineered commercial bike rack components using Onshape, optimizing geometry for manufacturability
- Validated system durability via hand-calculated static analysis, ensuring safety under peak physical load conditions
- Iterated designs alongside senior engineer, translating data into real-world manufacturing optimizations

PROJECTS

TTM Sponsored Senior Capstone Project

Fall 2025 – Present

- Designed an Industry 4.0 computer vision system for TTM Technologies to upgrade obsolete production assets
- Engineered components via CAD and fabricated a custom electrical control enclosure for seamless integration
- Developed an HMI using Ignition SCADA and used machine vision to streamline circuit registration
- Targeting a +/- 0.1 mil measurement accuracy, utilizing MySQL for automated data logging and analysis

Autonomous Obstacle-Avoiding "Smart Car"

Fall 2024

- Developed an autonomous vehicle using an Arduino Uno to govern kinematics and dynamic maneuvering
- Integrated distance sensors to detect obstacles, triggering audible alerts and visual indicators to signal detection
- Engineered collision-avoidance algorithms to evaluate sensor data and navigate complex obstacle courses

Ultralight Multifunctional Bike Wrench

Fall 2024

- Designed an aluminum multi-tool in SolidWorks, executing 8 geometric iterations to meet a 35.4g mass limit
- Validated structural durability via Ansys FEA and hand calculations for a 170 in-lb load, achieving a 2.79 FOS
- Fabricated the wrench from 6061-T6 aluminum stock and verified theoretical performance through conducted load testing

SKILLS & CERTIFICATIONS

CWSA Cert. (SolidWorks), Ansys, MATLAB, LaTeX, CNC, 3D Printing, MySQL, Ignition SCADA, FEA, Altair, Static Analysis, Dynamic Analysis, Arduino, Machine Shop Fabrication, Mechanical Load Testing, Lagrangian Mechanics, Industry 4.0